INCH-POUND
MIL-PRF-32043 (USAF)
26 April 1999

PERFORMANCE SPECIFICATION

INFLATION DEVICE, AUTOMATIC - FLU-9A/P

This specification is approved for use by the OO-ALC/TIEDE, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the FLU-9A/P automatic inflation device for use with the LPU-9/P automatic life preserver. The FLU-9A/P device shall provide the following: a storage container or cylinder for the inflating media, an actuator to release the media which shall have both a manual mode and an automatic mode initiated by immersion in water, a sensor to detect water immersion, and a port for connection to the LPU-9/P life preserver.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: OO-ALC/TIEDE, 6032 Fir Avenue, Hill AFB UT 84056-5820, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

BB-A-1034 - Compressed Air, Breathing

DEPARTMENT OF DEFENSE

MIL-PRF-25369 - Cartridge, Inflator, Flotation Gear, (Carbon Dioxide), (For the Inflation of Pneumatic Life Preservers)

(Unless otherwise indicated, copies of the above specification and standards are available from the Defense Automated Printing Service, Bldg 4D, (DPM-DODSSP), 700 Robbins Avenue, Philadelphia PA 19111-5094.)

2.2.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

LAWS AND REGULATIONS

40 CFR - Protection of Environment (Parts 82 and 302)

DRAWINGS

Air Force 9094276 - Lanyard Assembly
Air Force 9094296 - Configuration Item Interface, LPU-9/P
Air Force 9094354 - Life Preserver, Aircrew, Pneumatic, Automatic, Type LPU-9/P

(Copies of drawings required by contractors should be obtained from the procuring activity or as directed by the procuring activity.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD-adopted)

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(Application for copies should be addressed to American Society for Quality Control, 611 East Wisconsin Avenue, P.O. Box 3005, Milwaukee, WI 53201-3005, or to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

RADIO TECHNICAL COMMISSION FOR AERONAUTICS (RTCA)

RTCA/DO-160 - Environmental Conditions and Test Procedures for Airborne Equipment

(Application for copies should be addressed to the Radio Technical Commission for Aeronautics, 1140 Connecticut Ave, NW, Suite 020, Washington DC 20036-4001.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM – E1742 - Standard Practice for Radiographic Examination

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

3.3 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition, salt spray, and any other atmospheric condition that may be encountered during operational use or storage. Materials shall be compatible with any combustion products to which the device will be exposed. Materials that are nutrients for fungi shall not be used.

3.4 Pollution prevention. The inflation device, including all parts and equipment, shall not contain nor require for normal operation, maintenance, repair, or overhaul, any hazardous material unless approved by the procuring activity. A hazardous material is defined as a hazardous substance (40 CFR Part 302, Table 302.4) or an ozone depleting substance (40 CFR Part 82, Subpart A, Appendices A and B).

3.5 Interface.

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3.5.1 Container release assembly. The inflation device shall have a container release mechanism that will interface with the slide fastener on the LPU-9/P life preserver (see drawings 9094296 and 9094354). Upon activation, the inflation device shall release the slide fastener on the LPU-9/P and allow inflation of the life preserver.

3.5.2 Compatibility. The inflation device shall be compatible with the LPU-9/P. The device shall install on either the left or right lobe of the LPU-9/P flotation cell.

3.5.3 Lanyard assembly. The lanyard assembly shall be dimensionally in accordance with USAF drawing 9094276. The lanyard assembly shall withstand a load of 70 pounds.

3.6 Design.

3.6.1 Cartridges. Cartridges, if used, shall be approved by the procuring activity.

3.6.2 Gas cylinder. A gas cylinder, if used, shall conform to MIL-PRF-25369, Type II, carbon dioxide cylinder.

3.6.3 Electrical power source. The inflation device, if it uses an electrical circuit requiring a power source, such as batteries, shall not have the power source as part of the circuit until water immersion occurs. If the inflation device utilizes batteries, it shall have a provision for checking the voltage and polarity without any disassembly of the device.

3.6.4 Replaceable components. The following components of the inflation device shall be replaceable:

a. Carbon dioxide cylinder (if used)

b. Lanyard assembly

c. Batteries (if used)

d. Actuator safety restraint (see 3.7.2)

3.6.5 Size. The maximum dimensions of the inflation device shall be 7.6 inches for the length, 2.5 inches for the width, and 1.0 inch for the depth.

3.6.6 Weight. The maximum weight of the inflation device, less replaceable components, shall be 150 grams.

3.6.7 Structural integrity. The inflation device shall be free of defects such as cracks, voids, occlusions, foreign matter, and other flaws such as burrs, sharp edges, dents, and corrosion.

3.6.8 Useful life. The inflation device shall have a useful life (any combination of storage and service life) of a minimum of five years.

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3.6.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Performance.

3.7.1 Automatic system operation. The inflation device, installed in the LPU-9/P life preserver, shall automatically inflate the life preserver to its design shape within a maximum of 5 seconds when immersed in water of 67 to 70 micro-mhos/cm conductivity and a temperature of 70 ±4°F.

3.7.1.1 Automatic operation at temperature. The inflation device shall operate and release the inflating medium through the outlet port within a maximum of 5 seconds when immersed in water at a temperature of 32 to 36°F and 67 to 70 micro-mhos/cm conductivity, with the device temperatures stabilized at 70, -20, and 120°F (±4°F for each).

3.7.1.2 Automatic operation indicator. Automatic operation of the device shall cause a visual indicator of operation to be displayed.

3.7.2 Manual system operation. The inflation device shall contain an actuator for manually releasing the inflating medium. The device, upon manual actuation, shall inflate the LPU-9/P life preserver to its design shape within a maximum of 5 seconds. A replaceable safety restraint on the actuator shall prevent accidental operation. The device shall operate when a load greater than 5 pounds and not exceeding 20 pounds is applied to the actuator end.

3.7.3 Leakage. The inflation device shall not leak during full manual actuation with the outlet pressurized at 2 psig minimum. The device shall not leak with the outlet pressurized at 40 psig minimum and the manual actuator in the actuated position.

3.7.4 Electrostatic discharge. The inflation device shall not operate and shall not be damaged when subjected to a 25,000 ±500 volt electrostatic discharge.

3.7.5 Low-conductivity water resistance. The inflation device shall not operate when immersed in water at 70 ±4°F and 10 to 15 micro-mhos/cm electrical conductivity.

3.7.6 Electromagnetic radiation. The inflation device shall not operate and shall not be damaged when exposed to a radio frequency (RF) field at a strength level of 200 volt/meter, continuous wave, and a frequency range of 14 kHz to 18 gHz. A safety margin of 20 dB shall be required if electrically initiated cartridges are used.

3.7.7 Locked-shut mode. Locked-shut is defined as the condition where the actuating mechanism fails to move and the ballistic gas pressure is contained within the device. The inflation device, if it uses a cartridge, shall withstand the ballistic gas pressure when the cartridge is activated and the actuating mechanism fails to move while the inflation device is stabilized at a temperature range of -20 to +120 ±4°F.

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3.7.8 No-load operation. The inflation device, if it uses a gas cylinder, shall not be damaged upon automatic operation of the device with the gas cylinder removed, and the device stabilized at a temperature range of –20 to +120 ±4°F.

3.7.9 Hydrostatic pressure. The inflation device shall withstand a minimum hydrostatic pressure of 1,500 psig for a minimum of 5 minutes without damage or leakage.

3.7.10 Reliability. The reliability of the inflation device shall be 0.99 at 90% confidence level. If the inflation device contains an explosive cartridge, its performance parameters required to initiate actuation of the inflation device shall be identified to the procuring activity. The cartridge shall have a demonstrated reliability of at least 0.999 at 90% confidence level in achieving these parameters upon cartridge actuation.

3.7.11 Maintainability. The inflation device, if it uses a non-replaceable cartridge or other similar consumable, will be a one-time use item following automatic operation. Otherwise, the maintenance time to return the inflation device to a fully operational condition shall not exceed 15 minutes.

3.8 Identification. Each inflation device shall be permanently marked with the information below. The marking shall be placed so that it is visible with the device installed on the flotation cell.

INFLATION DEVICE, AUTOMATIC, FLU-9A/P SERIES

Manufacturer’s Identification
Contract or Order No.
Cartridge Lot No. (If a cartridge is used)
P/N & S/N
Lot No.
Month and year of manufacture
NSN

3.8.1 Warning label. A red, non-conductive label, with a white-lettered statement warning users about automatic operation upon water immersion, shall be permanently affixed to the inflation device. Wording shall be appropriate for the type of activating component used and shall be approved by the procuring activity.

3.9 Environmental conditions. The inflation device shall not automatically operate or show evidence of damage, such as cracks, deformation, or other flaws, during exposure to the conditions below, and shall function without degradation after exposure to the following conditions:

a. Rain of 350 micro-mhos/cm conductivity, 5 ±1 inches per hour, with 3500 ft per minute wind.

b. Salt fog.

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c. Dust.

d. Vibration.

e. Iced condition resulting from frozen condensation.

f. Storage for 360 hours at 160 ±4°F, and 48 hours at –65 ±4°F.

g. A drop of 6 feet onto a 2-inch thick steel plate.

h. A 20g shock load.

i. Temperature, humidity, and altitude cycling.

j. Windblast of 600 ±25 knots.

k. 23 full manual actuation cycles.

l. Explosive decompression.

3.10 Safety. Any hazard, known or potential, associated with the normal operation, maintenance, repair, or overhaul of the inflation device, or if any personnel protective equipment is required during such activity, shall be identified to the procuring activity.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the Section 3 requirements tested or verified in this section.

4.2 First article inspection. First article inspection shall consist of all examinations and tests described in 4.5. The test sequence is as shown in table II. Test quantities, unless modified by the procuring activity (see 6.2), are as shown in table II.

4.3 Conformance inspection. This inspection shall include the tests in 4.3.1 and 4.3.2.

4.3.1 Individual inspection. Each item shall be subjected to the tests below:

a. Examination (4.5.1).

b. Radiographic inspection (4.5.1.1).

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TABLE I. Requirements cross-reference matrix.

Requirement	Verification	Requirement	Verification
3.3	4.5.1, 4.5.14	3.7.1.2	4.5.2, 4.5.2.1
3.4	4.5.1	3.7.2	4.5.3
3.5.1	4.5.1	3.7.3	4.5.4
3.5.2	4.5.1	3.7.4	4.5.5
3.5.3	4.5.1, 4.5.11	3.7.5	4.5.6
3.6.1	4.5.1	3.7.6	4.5.7
3.6.2	4.5.1	3.7.7	4.5.8
3.6.3	4.5.1	3.7.8	4.5.9
3.6.4	4.5.1	3.7.9	4.5.10
3.6.5	4.5.1	3.7.10	4.5.12
3.6.6	4.5.1	3.7.11	4.5.13
3.6.7	4.5.1, 4.5.1.1	3.8	4.5.1
3.6.8	4.5.1.2	3.8.1	4.5.1
3.6.9	4.5.1	3.9	4.5.14
3.7.1	4.5.2	3.10	4.5.1
3.7.1.1	4.5.2.1		

c. Manual system operation (4.5.3a).

d. Leakage (4.5.4).

4.3.2 Sample tests. In addition to tests and examinations in 4.3.1, sample quantities shall be selected from a lot of completed inflation devices in accordance with inspection Level II of ANSI/ASQC Z1.4 and subjected to the tests and examinations below:

a. Automatic operation at temperature (4.5.2.1).

b. Manual system operation (4.5.3b).

c. Electrostatic discharge (4.5.5).

d. Low conductivity water resistance (4.5.6).

e. Hydrostatic pressure (4.5.10).

f. Lanyard tensile strength (4.5.11).

4.4 Inspection conditions.

4.4.1 Test items. Unless otherwise indicated in the test methods, all performance, environmental, and test requirements are based on an inflation device complete with replaceable components.

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TABLE II. Test sequence and quantities.

Test	Requirement	Test	Quantity
Visual inspection	3.3, 3.5, 3.6	4.5.1	204
Maintainability	3.7.11	4.5.13	10
Radiographic inspection	3.6.7	4.5.1.1	204
Automatic system operation	3.7.1	4.5.2	10
Automatic operation at temp	3.7.1.1	4.5.2.1	3 @ each temp
Manual system operation	3.7.2	4.5.3	10
Manual operation without cylinder	3.7.2	4.5.3a	204
Manual operation with cylinder	3.7.2	4.5.3b	10
Leakage	3.7.3	4.5.4	204
Reliability - inflation device	3.7.10	4.5.12	53 @ each temp
Reliability – cartridge	3.7.10	4.5.12	536 @ each temp
Electrostatic discharge	3.7.4	4.5.5	3
Low-conductivity water resistance	3.7.5	4.5.6	3
Hydrostatic pressure	3.7.9	4.5.10	10
Lanyard tensile strength	3.5.3	4.5.11	10
Temperature/humidity/altitude	3.9i	4.5.14.9	9
Storage	3.9f	4.5.14.6	2
Salt fog	3.9b	4.5.14.2	1
Vibration (unit at -20°F)	3.9d	4.5.14.4	1
(unit at 70°F)	3.9d	4.5.14.4	2
(unit at 120°F)	3.9d	4.5.14.4	1
Windblast	3.9j	4.5.14.10	2
Electromagnetic radiation	3.7.6	4.5.7	1
Manual cycling	3.9k	4.5.14.11	2
Dust	3.9c	4.5.14.3	1
Six - foot drop	3.9g	4.5.14.7	3
Shock	3.9h	4.5.14.8	1
Iced condition	3.9e	4.5.14.5	1
Rain	3.9a	4.5.14.1	2
No - load operation	3.7.8	4.5.9	2
Locked shut	3.7.7	4.5.8	2
Explosive decompression	3.9l	4.5.14.12	1

4.4.2 Test conditions. Unless otherwise indicated in the test methods, all required tests shall be conducted under the ambient conditions of pressure of 28 to 32 inches of mercury, temperature of 77 ±18°F, and relative humidity of 80% or less. If conditions change during a test, appropriate corrections shall be applied. Unless otherwise indicated in the test methods, test sample temperature conditioning shall be at least one hour longer than that required to reach thermal equilibrium.

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4.5 Tests.

4.5.1 Examination. The inflation device shall be examined to determine compliance with the requirements in 3.3, 3.4, 3.5.1, 3.5.2, 3.5.3, 3.6 (except sub-paragraph 3.6.8), 3.8, 3.8.1, and 3.10.

4.5.1.1 Radiographic inspection. The inflation device shall be subjected to radiographic inspection in accordance with ASTM E1742. The device shall show no evidence of cracks, voids, occlusions, missing parts, or misassembled parts.

4.5.1.2 Useful life. Unless waived (see 6.2), an inflation device with a date of manufacture at least 5 years prior to the test, shall be subjected to the tests required in 4.3.

4.5.2 Automatic system operation. The inflation device shall be assembled to the LPU-9/P life preserver and the LPU-9/P shall then be installed in a suitable fixture simulating service use. The complete unit shall be submerged in water having an electrical conductivity of 67 to 70 micro-mhos/cm at 70 ±4 ^{o}F. The inflation device shall operate and inflate the life preserver to its design shape within a maximum of 5 seconds. The automatic operation indicator shall be displayed.

4.5.2.1 Automatic operation at temperature. The inflation device shall be tested in water of 67 to 70 micro-mhos/cm conductivity. A different device shall be conditioned at each of the three temperatures of 70, -20, and 120°F (±4°F for each). The inflation devices, after each conditioning and within 3 minutes of removal from the chamber, shall be immersed in water at 32 to 36°F. The devices shall operate and release the inflating medium through the outlet port within a maximum of 5 seconds. The automatic operation indicator shall be displayed.

4.5.3 Manual system operation. The inflation device shall be assembled to the LPU-9/P life preserver and the LPU-9/P shall then be installed in a suitable fixture simulating service use and tested as follows:

a. The inflation device shall have a 5-pound load applied, in the operational direction, to the unsecured (no safety restraint) actuator end. The device shall not operate. The actuator mechanism shall not move from its initial position. This test shall be conducted with gas cylinder removed (if used).

b. The inflation device shall have a 20-pound load applied, in the operational direction, to the unsecured (no safety restraint) end of the actuator. The inflating medium shall be discharged by releasing 20-pound weights attached to the manual actuator on each device. The inflation device shall operate and inflate the life preserver to its design shape within a maximum of 5 seconds. The actuator shall then be secured with the safety restraint and the 20-pound load test repeated. The safety restraint shall not prevent the actuator from operating.

4.5.4 Leakage. An inflation device, with power source removed (to preclude automatic operation) and any gas cylinder port plugged, shall have a minimum of 40 psig applied at the outlet port. The device, with the actuator in the actuated position, shall then be immersed in

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water. There shall be no leakage. A minimum pressure of 2 psig shall then be applied to the outlet of the device with the manual actuator in the normal (unactuated) position and the device immersed. The manual actuator shall be slowly moved through at least one complete actuating cycle (normal to full actuated and back to normal position). There shall be no leakage at any point in the cycle. Compressed air shall be in accordance with BB-A-1034, grade A.

4.5.5 Electrostatic discharge. The inflation device shall be subjected to the potential of a 500 ±25 picofarad capacitor charged to 25000 ±500 volts discharged through a 5000 ±250 ohm resistor connected in series between the water sensing contacts.. The test points shall be reversed to change polarity and the test repeated. There shall be no damage or automatic operation. The device shall then be subjected to the tests specified in 4.5.2.1.

4.5.6 Low-conductivity water resistance. The inflation device shall be immersed in water at 70 ±4°F and 10 to 15 micro-mhos/cm electrical conductivity for a minimum of 60 seconds. The device shall not operate. The device shall then be subjected to the tests specified in 4.5.2.1.

4.5.7 Electromagnetic radiation. The device shall be tested in accordance with RTCA/DO-160, Section 20, Category Y (radiated susceptibility test level chart), with the following modifications:

a. Frequency range shall be from 14 kHz to 18 gHz, continuous wave field.

b. Electric field sensors are required from 14 kHz to 1 gHz. Either field sensors or receive antennas may be used above 1 gHz. Sensors shall be positioned 1 meter (±5%) from, and directly opposite, the transmit antenna.

4.5.7.1 There shall be no damage or automatic operation during the test. The device shall then be subjected to the test specified in 4.5.2.1, 70°F test only, with the water at 70 ±4°F. Safety margins, if cartridges are used, shall be demonstrated through instrumentation or by the use of cartridges rated 20 dB more sensitive but having the same thermal characteristics.

4.5.8 Locked-shut mode. If a cartridge is used in the design, two inflation devices shall be conditioned for a minimum of 4 hours, one at –20 ±4°F and the other at 120 ±4°F. Immediately after the conditioning, the actuating mechanisms shall be locked shut and the units operated by immersion in water of 50,000 micro-mhos/cm conductivity at 77 ±9°F. There shall be no rupture or mechanical failure.

4.5.9 No-load operation. This test is the same as the locked-shut test, except the actuating mechanism shall not be locked shut and the gas cylinder shall not be installed. There shall be no fragmentation of components and the actuating mechanism shall not be damaged.

4.5.10 Hydrostatic pressure. The inflation device, with the manual actuator in the actuated position and with the gas cylinder port(s) plugged (if applicable), shall have a minimum hydrostatic pressure of 1500 psig applied to its outlet port for a minimum of 5 minutes. The inflation device shall not show evidence of damage or leakage.

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4.5.11 Lanyard tensile strength. The tensile strength of the lanyard shall be verified by securing the manual actuator in a suitable fixture with a movable jaw, with the lanyard knob end attached to the movable jaw. A 70 pound load shall be applied to the lanyard. The lanyard assembly shall not break or fail.

4.5.12 Reliability test. Unless waived by the procuring activity (see 6.2), the reliability of the inflation device shall be demonstrated using the test in 4.5.2.1. Test quantities shall be as specified in 4.2.

4.5.13 Maintainability. Inflation device maintainability shall be verified by removal and replacement of the replaceable components (except for the lanyard) and by checking the voltage and polarity of the power source (if used) without requiring disassembly of the inflation device. The time for removal and replacement of all replaceable components shall be a maximum of 15 minutes.

4.5.14 Environmental conditions. Unless otherwise specified, the inflation device, after each environmental test, shall be inspected for damage and then subjected to the automatic operation test described in 4.5.2.1, 70°F test only, with the water at 70 ±4°F. After the vibration, temperature-altitude cycling, 6-foot drop, shock, and manual cycling tests, a radiographic inspection (see 4.5.1.1) shall also be conducted prior to the automatic operation test.

4.5.14.1 Rain. The inflation device, attached to the LPU-9/P life preserver assembly, shall be subjected to a rain test as outlined below. The inflation device shall then be subjected to the test described in 4.5.2.

a. Rainwater shall have a conductivity of 350 ±35 micro-mhos/cm at a temperature between 52 and 95°F. Diameter of water droplets shall be between 0.5 and 4.5 mm. Inflation device temperature shall be at least 15°F higher than the water temperature at start of test.

b. The test assembly, oriented for maximum exposure of the inflation device to the rain, shall be exposed to a simulated rain of 2 ±0.5 inches per hour for 10 minutes, increased to 5 ±1 inches per hour for 5 minutes, and then reduced to 2 ±0.5 inches per hour for the next 15 minutes. A horizontal wind of a minimum 3500 ft/min velocity shall be applied at 5 minutes after rain initiation and sustained for 15 minutes. Wind velocity shall be measured at the position of the test item.

4.5.14.2 Salt fog. The inflation device, less power source (if used), shall be subjected to the salt fog test in accordance with RTCA/DO-160, Section 14, Category S. The test duration shall be 48 hours.

4.5.14.3 Dust. The test shall be performed in accordance with RTCA/DO-160, Section 12, Category D, except the air velocity shall be 1750 ±250 ft/min and the dust concentration shall be 0.3 ±0.2 g/ft^3. The outlet port shall be protected to prevent clogging. Health and safety regulations should be observed if silica flour is used.

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4.5.14.4 Vibration. The inflation device shall be tested in accordance with RTCA/DO-160, Section 8, Standard and Random Vibration, Test Curve C, except the minimum acceleration power spectral density shall be 0.10 g^2/Hz minimum over the frequency spectrum from 300 to 1000 Hz. The vibration time per each axis shall be a minimum of 3 hours at each temperature of -20, 70, and 120°F.

4.5.14.5 Iced condition. The inflation device shall be conditioned at -20°F. It shall then be transferred to an atmosphere of 100°F and 90% relative humidity until all evidence of frost has disappeared. Retaining all condensation practicable, the device shall again be conditioned at -20°F for a minimum of one hour. After the conditioning, the inflation device shall then be tested as specified in 4.5.2.1 (-20°F only).

4.5.14.6 Storage. Two inflation devices, with the actuators unsecured, shall be used for this test. One shall be subjected to 160 ±4°F for 360 hours. The other device shall be subjected to -65 ±4°F for 48 hours. The inflation devices shall then be stabilized at a temperature of 70 ±4°F for a minimum of 2 hours before inspection and test (see 4.5.14).

4.5.14.7 Six-foot drop. Three inflation devices shall be dropped from a minimum height of 6 feet onto a 2 inch minimum thickness steel plate embedded in concrete to impact as follows: one with cartridge (if used) nose up, one with nose down, and one horizontal. A new device shall be used for each drop.

4.5.14.8 Shock. The test shall be conducted in accordance with RTCA/DO-160, Section 7. The operational shock test shall be applied, except the half-sine wave shall have an acceleration peak value of 20g. The total time duration shall be 6 to 9 milliseconds, with maximum acceleration occurring at approximately the midpoint of the time duration.

4.5.14.9 Temperature, humidity, and altitude cycling. The inflation device shall be tested in accordance with the test cycle in table III. Four complete cycles shall be required. There shall be no interim withdrawals of the device between the cycles. After completion of the four cycles, the inflation device shall then be inspected and tested (see 4.5.14).

4.5.14.10 Windblast. The LPU-9/P life preserver with the inflation device installed shall be secured to a PCU-15/P torso harness, or equivalent, and mounted on a 95th percentile mannequin. In addition, personal flight equipment, such as a survival vest, flight suit, helmet with visor, and oxygen mask, shall be mounted on the mannequin. The mannequin shall then be properly restrained in the aircraft ejection seat, or equivalent, and exposed to a 600 ±25 knots windblast. Rise time to peak velocity shall be 0.120 to 0.180 second, with no dwell at maximum velocity. Seat attitude shall be head-on and 34° pitched aft. The LPU-9/P assembly shall then be tested in accordance with 4.5.2.

4.5.14.11 Manual cycling. The inflation device, with the gas cylinder (if used) removed, shall be cycled through the manual mode of operation (actuator in normal position to full actuation and back to normal) for a total of 23 cycles. The inflation device shall then be inspected and tested (see 4.5.14).

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TABLE III. Temperature, humidity, and altitude test cycle.

Test Sequence	Chamber Temp, °F	Rel. Humidity, %	Chamber Press, psia[1]	Condition time, hrs[2]
High temp.	120	95	Amb.	4
IRP[3] & cold soak	-20	-	0.65 (70K ft. alt.)	16
Remove to	70	50	Amb.	4
IRP & cold soak	-20	-	0.65[4]	4
IRP & heat soak	120	95	Amb.	16
Reduce to	70	50	Amb.	4
Raise to	120	95	Amb.	4
IRP & cold soak	-20	-	0.65	16
Remove to	70	50	Amb.	4
IRP & cold soak	-20	-	0.65	4
IRP & heat soak	120	95	Amb.	16
Reduce to	70	50	Amb.	4
Raise to	120	95	Amb.	4
IRP & cold soak	-20	-	Amb.	16
NOTES:	1. Chamber pressure will be atmospheric when item is placed in chamber.			
	2. Condition time is after chamber temperature and pressure are stabilized.			
	3. IRP – Immediate removal and placement from one chamber to the other.			
	4. Time required to return pressure to 0.65 psia shall not exceed 1 hour.			

4.5.14.12 Explosive decompression. The inflation device shall be subjected to an explosive decompression from an altitude of 22000 feet to an altitude of 50000 feet (corresponding to pressures of 12.65 and 3.44 inches of mercury, respectively). Complete decompression or change of pressure shall take place in 0.1 second or less. The inflation device shall not operate and shall not be damaged nor have loss of operability.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The FLU-9A/P is a military service unique automatic inflation device intended for use with the LPU-9/P automatic life preserver. It can be operated manually by pulling the lanyard assembly or automatically by immersion in fresh or salt water. When activated, it sequentially releases the LPU-9/P container slide fastener and initiates inflation of the life preserver, which then inflates out of the container.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of the DoDISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. If the test quantities are different than in table II (see 4.2).

d. Whether a first article inspection is required (see 3.1 and 4.2).

e. If the service life test is waived (see 4.5.1.2).

f. Packaging requirements (see 5.1).

6.3 Cartridge qualification. Historically, cartridges used in the automatic inflation device have been subjected to service release test requirements as specified in MIL-D-21625, except for temperature and humidity cycling, salt fog, operation at 70°F, and operation at 80000 ft at -65°F.

6.4 Toxicology review. Any chemical or material required for the normal operation, maintenance, repair, or overhaul of the inflation device, which would require a Material Safety Data Sheet (MSDS), should be approved by the procuring activity. A complete MSDS should be provided for each chemical or material that requires an MSDS. If there is trade secret or proprietary information not releasable to the general public in the MSDS, the procuring activity should designate an environmental and occupational focal point, who under the Non-Release Clause will accomplish a toxicology review for submission to the procuring activity for approval.

6.5 Subject term (key word) listing:

Actuator

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Carbon dioxide
Cartridge, explosive
Cycling
Environmental
Hydrostatic
Immersion
Lanyard
Life preserver
Manual
Radiographic
Silica flour
Shock
Vibration

Custodian:
Air Force - 99

Preparing activity:
Air Force - 70

Review activity:
Air Force - 11

Project No. 1377-0099

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-32043	**2. DOCUMENT DATE (YYMMDD)** 99/04/26

3. DOCUMENT TITLE
Inflation Device, Automatic, FLU-9A/P

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED (YYMMDD)**

8. PREPARING ACTIVITY

a. NAME OO-ALC/TIEDE	**b. TELEPHONE** *(Include Area Code)* (1) Commercial 801-777-5538 (2) DSN 777-5538
c. ADDRESS ***(Include Zip Code)*** 6032 Fir Avenue Hill AFB, UT 84056-5820	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 756-2340 DSN 289-2340**

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